Machine learning in battery management systems: a two-tier scoping review of 799 eligible studies with in-depth classification and validation audit

Hussain Touhid Siddiquee, Syeda Salsabil Islam Ariya, and Jasimul Islam Chowdhury

***Abstract*—Machine learning (ML) has moved from a promising addition to battery research to a load-bearing component of modern battery management systems (BMS), yet the validation practices behind the literature's accuracy claims have never been systematically measured. This paper presents a structured scoping review built on a two-tier evidence base: an *eligible corpus* of 799 peer-reviewed journal articles (2019–2026), selected by uniform rule-based criteria applied to every screened record with no function quotas, and an in-depth *classified subset* of 103 studies coded by management function, method family, chemistry, data source, and validation practice, extended by an 11-paper conference-path descriptive check. Within-decimal-share bibliometrics on the eligible corpus locate the field's center of gravity in state and health estimation (SOH 20%, SOC 17%, prognostics 13% of eligible records) and show charging, thermal, and balancing functions and vehicle- or fleet-level work relatively under-served for a transportation-electrification readership. Three weaknesses measured in the classified subset persist across functions. Only 9 of 103 studies demonstrate online or fleet-level operation, while 17 rely on within-cell or k-fold splits that leak trajectory information; benchmark reliance is narrow, with 23 papers resting on the same three public datasets; and the laboratory-to-field gap remains the field's defining challenge. The validation contrast between physics-informed/hybrid papers and purely data-driven peers is exploratory, not conclusive: under a protocol-only definition of strong validation the gap is 33% versus 52% of papers (Fisher p = 0.15), and even under the broad historical definition reported here it is not significant (54% versus 73%, odds ratio 0.43, 95% CI 0.16–1.14, p = 0.13) — physics priors do not demonstrably buy validation rigor, but neither is the converse established. The paper closes with a mapping of BMS estimation functions onto UN GTR 20, GB 38031, and ISO 26262 requirements, a descriptive conference-path check, a safety taxonomy for foundation-model components whose first LLM-based estimator is analyzed as such, and a twelve-item reporting checklist (BMS-ML-RC) for future machine-learning BMS studies.**

***Index Terms*—machine learning, battery management systems, lithium-ion batteries, state of charge, state of health, remaining useful life, deep learning, physics-informed neural networks, transfer learning, battery safety**

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The authors are with Leading University, Sylhet, Bangladesh. The complete evidentiary record of this review—classified corpus, screening provenance log, protocol audit, codebook, and the blind inter-coder reliability study, together with the full analysis pipeline—is released as a Zenodo deposit, DOI: 10.5281/zenodo.23436202; see Data and Code Availability. Item 12 of the reporting checklist this paper proposes makes such a DOI-backed deposit the reporting norm.

I. Introduction

A. The battery management problem

Lithium-ion batteries now anchor three converging transitions — electrified road transport, renewable-heavy stationary storage, and the electrification of aerospace and marine systems — and in each the battery is an actively managed asset whose usable capacity, lifetime, and safety depend on real-time operating decisions. The battery management system (BMS) is the embedded cyber-physical layer that makes those decisions — sensing, state estimation, protection, balancing, and charging negotiation [1]. The difficulty stems from the fact that the quantities that matter most — state of charge (SOC), state of health (SOH), internal temperature, and the onset of degradation or fault mechanisms — are not directly measurable and must be inferred from noisy, partially observable terminal measurements under widely varying conditions. Model-based approaches built on equivalent-circuit or electrochemical models face a persistent fidelity-versus-tractability trade-off, and their parameters drift as cells age [2]. It is in this inference-and-decision layer that machine learning has made its most forceful entry, and nowhere with more at stake than in transportation electrification: electric-vehicle packs face automotive safety expectations — UN GTR 20 on board energy storage rechargeable systems, GB 38031 (GB/T 38031) on traction battery safety requirements, and ISO 26262 on functional safety of electric/electronic systems — that make state estimation, degradation prognosis, and fault detection safety-relevant BMS functions, not research add-ons. Section IX returns to these three standards and maps each estimation function to the property the standard actually demands (detection reliability, gradient/temperature limits, protection timing) — the mapping that turns the validation findings of this review into vehicle-level stakes. Relative to a general energy-systems readership, this framing also exposes the field's fit to this journal: laboratory cell-level estimation dominates the literature while pack-, vehicle-, and fleet-level practice remains thin, and the field's newest direction — vehicle-cloud orchestration — is itself automotive.

Machine learning entered battery management through estimation on hand-engineered features. Two studies then reset expectations. Severson and colleagues predicted the entire cycle life of lithium-ion cells with a median test error near 9% from the first hundred cycles alone, on a public dataset of 124 cells [3]. Attia and colleagues demonstrated closed-loop Bayesian optimization of fast-charging protocols in 2.5 months

rather than 18 months of manual experimentation [4]. The five years that followed produced applications across every BMS function: recurrent and convolutional networks for SOC and SOH [5] [6], graph and transformer architectures for prognostics [7] [8], physics-informed neural networks embedding electrochemical and thermal priors [9] [10], transfer learning across chemistries and vehicles [11] [12], reinforcement learning agents [13], and, most recently, large language model (LLM) agents for fleet-scale risk analysis and vehicle-cloud management [14] [15].

B. Existing reviews and the literature gap

The review literature has grown alongside the applications but is fragmented along the same axes as the primary studies: function-specific reviews cover SOC estimation [16] [17], SOH estimation [18], and prognostics [19]; method-specific reviews follow deep-learning state estimation [20], physics-informed impedance diagnostics [21], or interpretable prognosis [22]; and domain reviews examine fault diagnosis [23] [24] and thermal management [25], alongside broader degradation-method surveys [26] and EV-powertrain ML reviews with BMS scope [27]. What is missing is a cross-functional account of the field's trajectory: which method families are used for which tasks, on which data, validated how — and, among existing cross-functional surveys, none quantifies validation practice at all. Table VIII in Section IX positions this review against its closest competitors and states the added value explicitly; the short version is that prior reviews synthesize *what is claimed*, while this review additionally measures *how it is earned*, ties both to three safety standards, and releases the record-level dataset to make every count auditable.

C. Contribution of this paper

This paper makes four contributions: (i) a rule-based *eligible corpus* of 799 peer-reviewed journal articles (2019–2026), selected by uniform criteria applied to all screened Crossref records — no function quotas, year caps, or citation thresholds — so that field-level bibliometrics are measured rather than selected into being; (ii) an in-depth *classified subset* of 103 studies from that corpus, coded along eight axes (function, method family, chemistry, data source, validation practice, and more) with the classification dataset, codebook, per-paper protocol audit, and 30-paper inter-coder reliability study released; (iii) a two-tier synthesis — measured bibliometrics from the eligible corpus (Section III) plus function-by-function critical review of the classified subset (Sections IV–VIII) — with a conference-path descriptive check, a BMS-function-to-safety-standard mapping, and a gap-driven research agenda including a twelve-item reporting checklist (Section IX); and (iv) a transportation-electrification cut of the corpus (Section III-B) separating cell-level laboratory evidence from pack-, vehicle-, and fleet-level practice. Fig. 1 previews the classified subset, mapping it across management functions and method families: the center of gravity sits in the health-prognostics complex, hybrid model–data approaches span every function, and learning-based charging control remains a concentrated Bayesian-optimization stronghold. The remainder of the paper is organized as follows. Section II details the review methodology and its limitations. Section III presents the two-tier corpus overview, five-year bibliometrics, and the transportation-electrification cut. Sections IV–VIII review machine learning for state estimation, health estimation and prognostics, charging control, thermal management, and fault detection and safety. Section 8 examines cross-cutting directions, and Sections IX and X give the research agenda and conclusions.

II. Review Methodology

A. Search strategy

All candidate records were retrieved programmatically from the Crossref REST API between 4 and 5 September 2026 using 30 bibliographic query strings spanning the application areas above. Queries were issued in four passes (top-cited 2020–2026; recency from mid-2024; dedicated 2020–2022 and 2023 windows), yielding 3,000 records; after DOI-based deduplication (1,450 duplicates removed), 1,550 unique records remained. Six landmark studies — including [3] and [2] — were additionally safeguarded as forced seeds: five were retrieved and verified individually because citation-sorted retrieval had crowded them out, and one was already among the top-cited hits. Two properties should be stated plainly. The Crossref registry is a metadata registry, not a discovery engine like Scopus or Web of Science; it was chosen so that every included record is individually verifiable by direct DOI resolution, at the cost of a recall floor. Citation ordering also favors established English-language work; all trend statements therefore describe this citation-weighted, journal-article sample rather than the field in the aggregate.

B. Screening and rule-based selection

The screening funnel is shown in Fig. 2. A title-level relevance gate requiring both battery-related and machine-learning terminology retained 883 records and excluded 667 (predominantly adjacent-field papers that surfaced through loose matching). Venue and document-type screening removed 84 records — conference abstracts and proceedings series, preprint servers, and non-scholarly venues — leaving 799 eligible records. Selection then proceeds in two tiers, replacing the single-quota design that an earlier draft of this review used. *Tier 1 (eligible corpus).* Every one of the 799 eligible records is included, subject only to venue screening; no function quotas, year caps, or citation thresholds are applied. Every one of the 799 eligible records carries a Crossref-registered DOI, and every record carried forward to Tier 2 full-text coding was individually DOI-resolved; every field-level distribution reported from this tier (function shares, publication years, venue concentration) is a measurement of the eligible literature, not an artifact of selection. *Tier 2 (classified subset).* Because in-depth coding of chemistry, data source, and validation practice requires full-text assessment, full-text coding was applied to a 103-study subset of the eligible corpus, curated with the same eight-function stratification used by the tagging pass, with the 2019 field-defining seed (Severson et al.) and six landmark studies ([4], [28], and four others) included as forced entries. This subset is curated, not sampled: it preserves a selectable

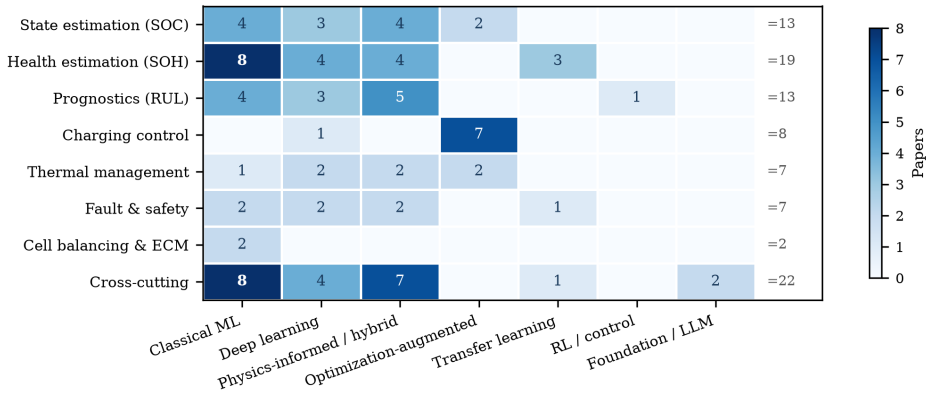


Fig. 1. The classified subset mapped across battery-management functions and method families (the 91 primary-application papers; the 12 method-coded reviews are excluded). Table II states the reconciliation: 12 reviews with method label "Review / perspective" plus 3 review-type syntheses coded under their dominant method family; no review belongs to the physics-informed-family total of 24 (Table I).

record-level rationale for every study included or dropped, released as a machine-readable supplement (screening_log.csv), and its function and year composition does not match the eligible corpus's measured shares exactly. Section III therefore never generalizes subset distributions to the field; where the subset is the only source (method family, validation practice), the text reports subset statistics and their selection caveats explicitly.

C. Classification framework, validation definitions, and limitations

Each paper was classified along six fields: management function (eight values, including a cross-cutting cluster); dominant method family (classical ML, deep learning, physics-informed/hybrid, optimization-augmented, reinforcement/control learning, transfer learning, foundation models/LLMs, review); chemistry; primary data source; validation approach; and a coded summary of the headline finding and principal limitation. Three terms are used consistently: an *eligible record* is one of the 799 Tier-1 studies; a *classifiable paper* is one of the 103 Tier-2 studies; a *primary-application paper* is one of the 91 classified papers whose contribution is a primary application rather than a review; and review-type papers are counted as reviews (12) wherever the term appears — three further review-oriented syntheses are coded under their dominant method family because they contribute classification frameworks rather than, or in addition to, narrative synthesis, and Table II labels each such paper explicitly. Validation practice is coded against four definitions used throughout this paper: (1) *within-cell split* — train and test segments drawn from the same cycling trajectory; (2) *unsupervised k-fold cross-validation* — folds assigned over samples of the same trajectories without cell isolation; (3) *cross-cell holdout* — every test cell held out entirely from training (with cross-chemistry and cross-domain holdout extending this isolation across chemistry or application); and (4) *online or real-vehicle/fleet validation* — the deployed system evaluated on live vehicle data. Cross-cell, cross-chemistry, cross-domain, and online/fleet are together the *strong protocols*; the four remaining coded environments (experimental hardware, simulation plus hardware spot-check, simulation-only, case study) characterize where a study was exercised but say nothing about train–test isolation, and they are reported separately — never mixed into strong-protocol percentages. Borderline assignments, the absorbing rule for ambiguous cases (a study is credited with the *strongest* protocol it reports for its headline claim), and the decision tree with examples are given in the released codebook. Classification was performed by the lead author with conservative coding from full texts; to measure the reliability of that coding, two independent coders later blind-re-coded a 30-paper stratified random sample (drawn by function and year band from the classified subset, seed-pinned and reproducible) under the released codebook, without access to the released labels, to the population statistics, or to each other. Blind agreement between the two independent coders is strong on three of the five classification fields: management function (κ = 0.92, Krippendorff α = 0.92), method family (κ = 0.88, α = 0.88), and primary data source (κ = 0.88, α = 0.88). Two fields fell below the $\kappa \geq 0.66$ threshold: chemistry (κ = 0.55) and validation protocol (κ = 0.48) — but the failure mode is informative rather than random. 23 and 28 of the 30 sample papers (coders 2 and 3, respectively) could be coded from abstracts alone, and the two fields are precisely those least often stated at abstract level: both independent coders resolved unreported split protocols toward weaker codes (so their agreement with the labels of coder 1 on isolated field values reflects evidence access, not label error), and on the aggregation that Section III's counts rest on — strong protocol versus non-strong — the two coders agree on 25 of 30 papers (83%). Per the pre-registered protocol, the two below-threshold fields trigger codebook repairs (an explicit validation code for "protocol not reported" and a chemistry inference rule for abstract-level coding), versioned as candidates without altering any released label; the complete per-paper, per-field disagreement table is released together with both locked coding sheets and the hash-tracked log, so any reader can re-audit or re-score the reliability claim. The synthesis proceeds on two tracks: measured bibliomet-

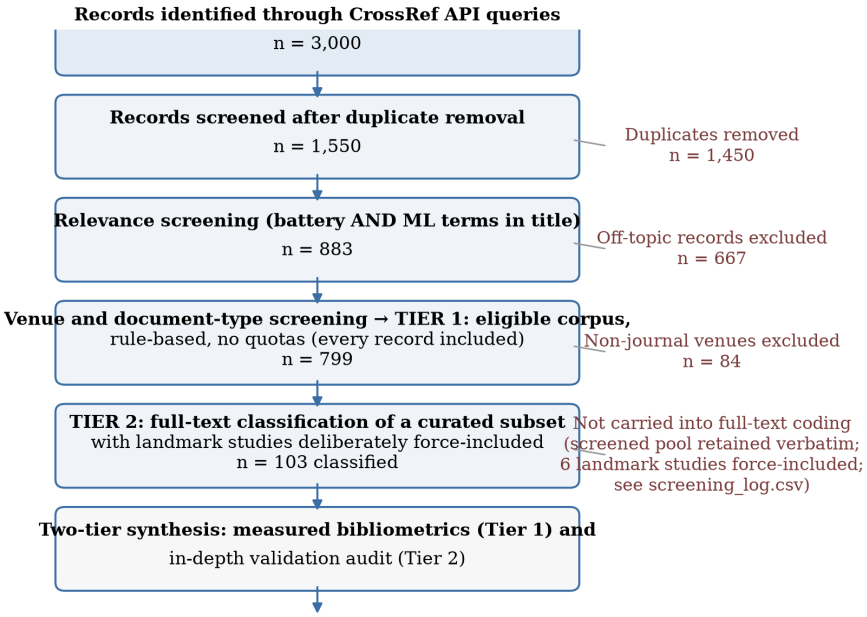


Fig. 2. Study-selection flowchart with the two-tier design. Records were retrieved from the Crossref REST API on 4–5 September 2026, deduplicated, screened at title level for battery and machine-learning relevance, screened at venue and document-type level. Tier 1 includes every eligible record (799, rule-based, no quotas); Tier 2 carries a curated full-text-classified subset of 103 with six landmark studies included deliberately and five force-included individually from outside the pool. Record-level provenance is released as screening_log.csv.

rics from the eligible corpus and cross-tabulations from the classified subset, both generated programmatically from the released datasets (Section III), and qualitative function-by-function critique (Sections IV–VIII). The methodology's limitations are acknowledged: Crossref coverage excludes influential conference papers; title-level screening can miss relevant work; citation-ordered retrieval privileges established English-language venues; the Tier-2 subset is curated rather than sampled; and reliability is documented on a 30-paper blind recoding subset rather than measured corpus-wide (three of five fields above threshold, two fields flagged for codebook repair). DOI resolution, moreover, verifies that a record exists, not that its venue is topically sensible: one corpus study (a physics-informed SOC network, [10]) appears in an RFID-special-issue journal — we flag it here rather than dropping it, and the released audit file (final_audit.json) records every reference's Crossref-resolved title and venue so that misattributions of this kind are checkable rather than hidden. These are the reasons this paper describes itself as a scoping review with bibliometric classification rather than a meta-analysis.

III. Corpus Overview and Five-Year Evolution

A. Publication trend and method-family evolution

Two layers of trend evidence are reported. On the *measured* tier (eligible corpus, 799 records), output rises essentially monotonically: 55, 85, 139, and 223 journal articles in 2020–2023, a retrieval-window dip to 56 in 2024 (records published in the most recent [≈24-month] horizon are incompletely captured by citation-ordered Crossref passes), then 126 and 114 in 2025–2026 (Fig. 3, left panel). Whatever the exact scale factor, the direction is unambiguous — the ML-BMS journal literature at least quadrupled between 2020 and 2023 — and its composition concentrates in estimation functions (next subsection). On the *classified subset*, the defining finding is the compositional shift in method families (Fig. 3, right panel; Table I). Classical ML — Gaussian process regression, support vector machines, ensembles, shallow networks on engineered features — contributes 11 of the 23 subset papers published from 2019 through 2021, while physics-informed and hybrid model–data learning rises year by year (1, 2, 3, 4, 5, 4, and 5 papers) to become the modal family by 2024–2026 — 14 of its 24 subset papers fall in 2024–2026, and it is the largest component of the 7 papers published in 2026 (Table II gives corpus-wide totals). Optimization-augmented learning (Bayesian optimization, surrogate-assisted and meta-heuristic schemes) peaks in 2023–2024 (11 papers); transfer learning accelerates in 2024–2025 (5 papers); reinforcement learning remains scarce (1 paper); the first foundation-model and LLM-based studies appear in 2025–2026 (2 papers) [14] [15]. These family shares describe the subset only; the subset is curated and small-denominator years (2025: 6, 2026: 7) carry wide sampling error. Two qualifications matter: the families are not mutually exclusive (a PSO-tuned LSTM is both deep learning and optimization-augmented), so the composition understates method mixing; and physics-informed papers are, on average, validated no more stringently than others even under the most

permissive definition of stringency — a point quantified, with confidence bounds, in Section 3.3.

B. Measured application areas, the transportation-electrification cut, and venues

Because function tags were applied uniformly to all 799 eligible records, the application-area distribution can be *measured* rather than quoted from a curated sample (a record can carry more than one tag; shares are per record tagged). Health estimation (SOH) leads at 20% of eligible records (158), followed by state estimation (SOC) at 17% (133), prognostics (RUL) at 13% (103), thermal management at 12% (99), fault and safety at 12% (96), charging control at 8% (64), and cell balancing at 5% (38) — the health-prognostics complex together with SOC occupies half of the eligible literature, a measured pattern consistent with the commercial pull of lifetime prediction for warranties and second-life markets, and mirrored (with a heavier cross-cutting share) in the classified subset (Fig. 1; Table II gives subset composition by year; Table V maps representative studies). Two measured concentrations matter specifically for this journal. First, the transportation-electrification cut: only ≈2% of eligible records ($n = 17$) are tagged fleet- or cloud-orchestration-adjacent, and in the classified subset only 10 papers use real vehicle or fleet data as their primary source against 36 resting on laboratory cells — the literature is overwhelmingly a cell-level, laboratory phenomenon, sitting one integration level below the pack- and vehicle-level questions that UN GTR 20, GB 38031, and ISO 26262 pose. Second, work explicitly aimed at onboard deployment is correspondingly scarce. The thin representation of cell balancing is also telling: balancing remains dominated by classical control practice, and the subset's most-cited balancing-related result is the demonstration that routine balancing can be dispensed with in well-managed packs, reducing hazardous lithium-plating side reactions [29]. Venue concentration is equally marked — in the eligible corpus the Journal of Energy Storage (103 records), Energy (61), and Batteries (45) lead, and the same three journals head the classified subset (14, 13, and 9 papers respectively) — energy-storage venues, not transportation-electrification venues, absorb the field's output.

C. Data landscape and validation practice

The data underpinning the classified subset are strikingly concentrated: 36 papers rest on laboratory cycling of the authors' own cells; the public Severson/Toyota–MIT–Stanford (MATR) dataset alone supports 11, NASA and CALCE 6 each, and laboratory impedance spectroscopy 6; 12 are simulation-based. Only 10 papers use real vehicle or fleet data as their primary source, and *data use* is a distinct measure from *validation protocol*: 8 of the 10 fleet-data papers are validated online or on real vehicles, 2 are validated instead by transfer or case-study protocols, and 1 further study achieves online validation on laboratory data (Table III). Validation practice is arguably the field's most consequential weakness, and it is now defined and audited explicitly. Per the definitions of Section II-C, cross-cell holdout — the strongest common laboratory protocol — accounts for 24 primary-application papers, cross-chemistry holdout for 7, cross-domain holdout for 3, and online or fleet validation for 9; together these *strong protocols* cover 43 of the 91 primary-application papers (47%). Within-cell splits (12 papers) and unsupervised k-fold cross-validation (5) leak trajectory information between train and test, producing optimistic errors that do not survive cell-to-cell variability; the remaining studies establish hardware operation (19), simulation- or case-level evidence without a generalization protocol (9), or are review-style syntheses coded under a primary method family (3) — 43+17+31 = 91. The complete per-paper protocol assignment is released in papers.csv (valid field) and tabulated as a supplement, so the 43-of-91 arithmetic above can be verified row by row.

Crossing validation practice with method family, the cautious reading is: 8 of the 24 physics-informed/hybrid papers (33%) use a strong protocol, against 35 of the 67 purely data-driven or other primary-application papers (52%) — Fisher's exact $p = 0.15$, odds ratio 0.46 (95% CI 0.17–1.21). Under the earlier, broader definition that also counted experimental-hardware demonstration as strong validation, the contrast is 13 of 24 (54%) versus 49 of 67 (73%), OR 0.43 (95% CI 0.16–1.14, $p = 0.13$). We report both and claim neither as a model-family effect: the difference is not statistically significant under either definition, confounding is unadjusted (physics-informed work skews toward early-stage and simulation-heavy studies), and $n = 24$ is small. The honest reading is that a physics prior does not *demonstrably* buy validation rigor — and that settling the question requires the standardized protocols proposed in Section IX, not more corpus cross-tabs.

The benchmark concentration can be stated as a metric (Fig. 4). Across 2019–2026, 23 corpus papers — more than a fifth of the primary-application corpus — rest on the same three public benchmarks. Because these benchmarks together contain on the order of a few hundred cells, the field's effective empirical sample has grown far more slowly than its publication count: output rose from single digits per year in 2019–2020 to 18–22 papers per year in 2023–2024, while benchmark reuse kept pace step for step. We call this pattern benchmark saturation: the empirical base on which the community calibrates and compares estimators is approximately fixed while the literature multiplies on top of it, inflating apparent comparability and hiding dataset-specific failure modes.

D. Conference-path descriptive check

Because the corpus excludes conference proceedings — a major venue for control-oriented BMS work (ACC, ECC, CCDC, APEC, VPPC, ITEC) — the conference path deserves its own look. We compiled an 11-paper descriptive set of *machine-learning* battery papers from six control and transportation-electrification venues (2019–2025), retrieved by venue-scoped Crossref queries, restricted to proceedings articles, ranked by citation count, and verified by the same DOI-resolution procedure as the main corpus (released as conference_sensitivity.csv). Entries whose headline method is conventional model-based estimation rather than learning (a dual extended Kalman filter, a batch-least-squares pack estimator, and an EKF critique) are excluded from this set

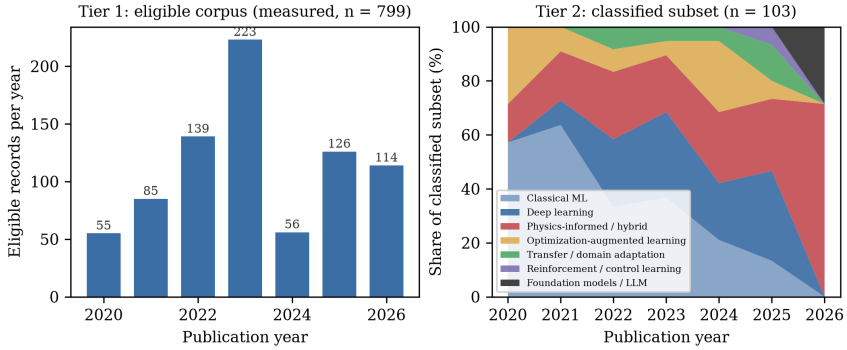


Fig. 3. Two-tier trend evidence. Left: Tier 1 eligible-corpus publications per year (measured across all 799 records; the 2024 dip reflects the citation-ordered retrieval window, not the field). Right: Tier 2 classified-subset method-family shares by year (curated subset, wide sampling error in small years; the 2019 seed is omitted for scale).

TABLE I
METHOD FAMILIES IN THE CORPUS: REPRESENTATIVE TECHNIQUES, STRENGTHS, AND LIMITATIONS

Method family	Papers	Representative techniques	Strengths	Limitations
Classical ML	29	Gaussian process regression, SVM/SVR, ensembles, shallow networks on engineered features	Calibrated uncertainty; small-data friendly; cheap to embed	Feature engineering is manual; ceilings under complex dynamics
Deep learning	19	LSTM/GRU, CNN, attention/Transformer, autoencoders, graph networks	Learns features from raw curves; strong trajectory modeling	Data-hungry; weak extrapolation; opaque
Physics-informed / hybrid	24	PINNs, physics-constrained autoencoders, filter-embedded networks, multiphysics fusion	Structure aids extrapolation; reduced data demand	Inherits simplified-model bias; harder to train
Optimization-augmented	11	Bayesian optimization, surrogate-assisted multi-objective design, meta-heuristic-tuned networks	Sample-efficient protocol discovery; explicit trade-off handling	Offline tuning cost; surrogate bias
Transfer / domain adaptation	5	Domain-adversarial training, fine-tuning across chemistry/protocol	Directly targets the generalization gap	Negative transfer; needs shared protocols
Reinforcement / control learning	1	Deep-RL prognostic model selection	Adaptive decision-making along trajectories	Training stability; scarce hardware validation
Foundation models / LLM	2	Domain-tuned LLM agents for risk analysis and orchestration	Reasoning over heterogeneous text/system context	Hallucination risk; latency and cost
Review / perspective	12	Narrative and systematic syntheses	Maps the field; identifies shared gaps	No primary validation

TABLE II
CORPUS COMPOSITION: MANAGEMENT FUNCTION BY PUBLICATION YEAR. THE REVIEWS COLUMN COUNTS THE 12 METHOD-FAMILY REVIEWS; PRIMARY = TOTAL − REVIEWS GIVES THE PRIMARY-APPLICATION COUNTS USED IN FIG. 1.

Function	'19	'20	'21	'22	'23	'24	'25	'26	Total	Reviews	Primary
Cell balancing & ECM	0	0	0	0	1	1	0	0	2	0	2
Charging control	0	2	0	0	2	3	1	0	8	0	8
Cross-cutting	0	1	0	6	8	3	2	4	24	2	22
Fault & safety	0	0	2	0	2	3	2	0	9	2	7
Health estimation (SOH)	0	2	3	2	4	3	4	2	20	1	19
Prognostics (RUL)	1	1	3	2	1	1	6	0	15	2	13
State estimation (SOC)	0	3	4	2	1	5	2	0	17	4	13
Thermal management	0	0	1	1	3	1	1	1	8	1	7
Total	1	9	13	13	22	20	18	7	103	12	91

TABLE III
CROSS-TABULATION OF FLEET-DATA USE AGAINST ONLINE VALIDATION (103 CLASSIFIED PAPERS). FLEET-DATA USE DESCRIBES THE PRIMARY DATA SOURCE; ONLINE VALIDATION DESCRIBES THE VALIDATION PROTOCOL; THEIR OVERLAP IS 8 OF 10 FLEET-DATA PAPERS. THE RIGHT-HAND COLUMN POOLS THE ENVIRONMENT-ONLY VALIDATION CATEGORIES (SECTION 3.3).

	Online / real-vehicle validation	All other validation protocols (Table 5)	Total papers
Real-vehicle / fleet data	8	2 (transfer; case study)	10
Other data sources	1 (laboratory-data demonstration)	92	93
Total	9	94	103

and shown instead in Section VII's discussion of filter–learner hybrids. Validation practice was coded conservatively from the indexed abstracts (Table IV). The result is descriptive, not a test of venue bias: only 2 of the 11 conference papers (18%)

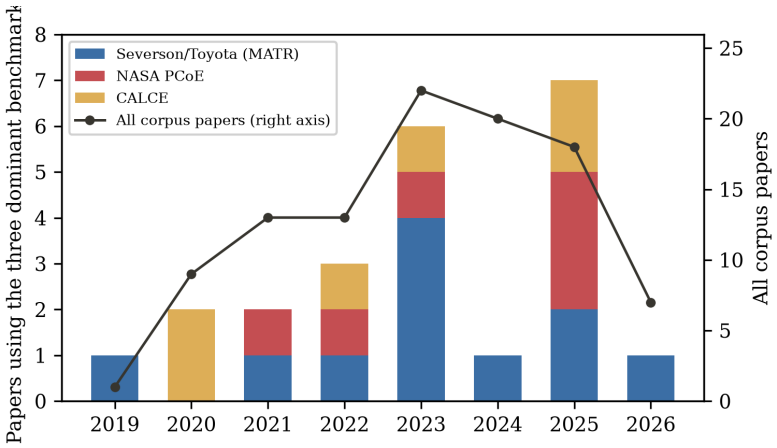


Fig. 4. Benchmark saturation. Papers using the three dominant public benchmarks per year (stacked bars, left axis) against total corpus output per year (line, right axis).

state hardware validation in their indexed abstracts, 2 rest on public datasets, and 3 describe simulation-level validation — against 19 of the 91 primary-application journal papers (21%) with experimental-hardware demonstrations and a 47% strong-protocol share. Three caveats bound any inference: conference abstracts are frequently truncated in bibliographic indexes, so the comparison measures stated evidence rather than underlying practice; the coverage bases of the two sets differ (journal subset papers were assessed full-text, conference papers from abstracts); and the set is a curated sample, not a census. Within those bounds we conclude only this: neither venue type reports validation systematically on its face, which strengthens the case for the reporting standards proposed in Section IX.

IV. Machine Learning for Battery State Estimation

SOC estimation is the most mature BMS function, and the corpus shows how machine learning entered it. The early pattern pairs learning with classical estimation: Gaussian process regression for pack-level SOC with native uncertainty quantification [41], and — most influentially — recurrent networks embedded in unscented Kalman filters, where the LSTM supplies a learned observation model and the filter recursive correction [5]. A second generation tuned network structure itself: particle-swarm-optimized LSTM improved accuracy and noise robustness [42], and evolutionary-assisted CNN–BiLSTM hybrids report sub-1% drive-cycle errors [43]. Simulation-informed training pipelines substitute automotive-accurate simulated data for exhaustive cell testing [44], comparative benchmarks on standard drive cycles weigh ML regression families [45], pack heterogeneity is addressed by joint pack-SOC and cell-inconsistency estimation with balancing-aware formulations [46], and unified frameworks estimate SOH and SOC jointly from partial charging data [47], and large-format applications extend estimation to containerized storage strings [48], with comparative benchmarks weighing accuracy against embedded compute budgets [49] [50].

The most substantive recent shift is toward hybrid schemes in which a filter, an equivalent-circuit model, or a physics prior constrains the learner: GA-optimized LSTM coupled with improved adaptive Kalman filtering reaches low error under measurement noise that breaks pure-network estimators [51]; equivalent-circuit-informed ML on impedance features estimates SOC across temperatures [52]; and physics-informed recurrent structures embed battery dynamics directly into the training objective [10]. These schemes trade engineering complexity for extrapolation behavior — the property that matters most onboard. Yet 13 of the 17 corpus SOC papers rest on laboratory-scale data and 12 on a single chemistry, and only one demonstrates fleet-scale operation [48]; benchmark heterogeneity makes cross-paper comparison nearly meaningless [20] [17]. The laboratory problem is largely solved; the deployment problem — guaranteed accuracy on arbitrary packs over full aging — is not.

V. Machine Learning for Health Estimation and Prognostics

SOH estimation and RUL prediction form the largest functional block of the corpus (35 papers). The founding pattern is classical: extract health-sensitive features from partial charge or discharge curves, then regress with Gaussian process or similar shallow models — incremental-capacity analysis with explicit uncertainty bounds [53], differential thermal voltammetry [54], and Gaussian-process classification of impedance spectra that identifies degradation modes [55], and deep Gaussian processes coupling prognosis with degradation-mode identification [56]. Measurement relaxations followed: random partial charging data proved sufficient for accurate SOH [57], and charging-segment adjustment refined features for vehicle-level use [58]. The landmark systematization is Roman et al.'s 179-cell pipeline showing that relaxed charge-curve features support accurate SOH estimation across a heterogeneous commercial fleet [28].

Deep architectures entered as sequence models over capacity-fade trajectories and representation learners over raw curves: graph convolutional networks fusing multiple health indicators [7], graph networks over partial discharge curves [59], BiLSTM–Transformer hybrids under fast-charging stress

TABLE IV
CONFERENCE-PATH DESCRIPTIVE SET: 11 MACHINE-LEARNING BATTERY PAPERS FROM CONTROL AND TRANSPORTATION-ELECTRIFICATION VENUES (2019–2025), CROSSREF-VERIFIED, WITH VALIDATION EVIDENCE STATED IN THEIR INDEXED ABSTRACTS.

Venue (year)	Function	Method family (short)	Validation evidence stated in indexed abstract
ITEC 2019	SOC	LSTM recurrent network [30]	Not stated
VPPC 2021	SOH	ML ensemble [31]	Public dataset
ACC 2021	SOC	Multiple-model adaptive estimation [32]	Simulation-level
ITEC 2023	SOH	Histogram data + PCA + regression [33]	Public dataset
APEC 2023	Overview	Review of ML-enabled estimation [34]	Not applicable
ECC 2021	SOH	Reinforcement learning + observer [35]	Not stated
CCDC 2022	SOH	CNN-BiLSTM transfer learning [36]	Hardware (stated); public data
VPPC 2019	SOC	SVR + PCA with dual-polarization model [37]	Not stated
VPPC 2022	Prognostics	ML on aging data [38]	Hardware (stated)
VPPC 2019	SOC	Kalman filter on reduced electrochemical model [39]	Simulation-level
VPPC 2021	SOC	Computationally efficient neural network [40]	Simulation-level

[8], hybrid-fusion networks for real-time prediction streams [60], and interpretable deep learning with transparency mechanisms [61]. Field-data studies absorb real-vehicle noise [6] and estimate pack-level SOH from fleet streams [62], while deep models estimate SOH without additional degradation experiments by learning trajectory features that generalize across chemistries [63]. Early-life prediction — forecasting lifetime from the first cycles — compresses test times from months to days [3] [64] [65], hybrid physics-plus-learned models give early RUL from limited cycles [66], extends to lithium-metal chemistries [67], and now runs on embedded boards in real time [68].

The prognostics cluster shows the clearest physics-informed turn: physics-informed online degradation models [69], hybrid reinforcement learning that reduces the data demand of SOH estimation [70], a generalizable PINN from limited early-cycle data [71], EIS-informed physics-constrained networks [72], a generic physics-informed RUL framework for small early-life samples [73], multi-feature hybrids across datasets [74], recurrent networks anticipating uncertain duty [75], and signal-decomposition hybrids on noisy capacity series [76]. Transfer learning attacks generalization directly: domain-adversarial training across operating conditions [11], deep transfer with partial charging curves across cell types [12], transfer under fast-charging protocols [77], cross-chemistry EIS estimation [78], and a general framework quantifying the laboratory-to-fleet domain gap [79]. The critical assessment: accuracy claims cluster around 1–2% SOH error, but they are established overwhelmingly on public benchmark cells — 10 of the 35 studies in this complex use within-cell or k-fold validation (17 corpus-wide) — and the transfer-learning papers, the honest exception, reveal how much shift remains unaddressed. SOH estimation under controlled conditions is approaching incremental returns; calibrated lifetime and safety margins for arbitrary packs in the field remain open [2] [19].

VI. MACHINE LEARNING FOR CHARGING CONTROL AND OPTIMIZATION

Charging control is the function where machine learning has changed experimental practice most directly. The defining result is Attia et al.: a Bayesian optimization loop identified fast-charging protocols within 2.5 months, against roughly 18 months of manual search [4]. Ensemble surrogates make multi-objective strategy optimization tractable [81], and Bayesian fast-charging optimization reduces charge time under degradation constraints [88]. A second line embeds degradation mechanisms into the decision itself: machine-learning fast charging that perceives and regulates internal lithium-plating onset enables plating-free rapid protocols [80], adaptive multi-stage schemes adjust stage boundaries against measured degradation [89], module-level optimization manages temperature gradients [90], and pack-level extensions coordinate balance-aware charging [91] [92].

The deployment gap is most concrete here: of the eight charging papers, one is a large-scale closed-loop hardware campaign, three validate on laboratory hardware, and the rest are simulation-only. The unsolved problem is certifiable learning-based charging — which is why mechanism-aware schemes such as plating-perception control [80] are the cluster's most deployment-relevant results.

VII. MACHINE LEARNING FOR THERMAL MANAGEMENT, FAULTS AND SAFETY

Thermal management (8 papers) divides by purpose. For estimation, ANN models predict cell temperature across operating conditions [82] and internal-temperature sensing extends to ultrahigh-rate discharge where surface measurement fails [93]. For design, ML surrogates amortize expensive CFD evaluation across optimization loops [94] [83], with experimentally validated parallel-liquid-cooling designs closing the loop [95]. For control, ANN controllers demonstrate parasitic-load reductions in simulation [96] — but no corpus paper closes this loop on hardware, leaving thermal control where charging control was five years ago. Physics-informed heat-generation estimation embeds thermal priors into the learning problem [9], and the cluster is consolidated by a state-of-the-art review [25].

Fault and safety applications (9 papers) divide into diagnosis of realized faults and prediction of impending hazards: grid-search-optimized SVMs classify sensor and connection faults [84], curvilinear-distance methods detect multiple pack fault types in operating fleets [97], and multi-scenario deep anomaly detection covers real EV operating envelopes [98];

TABLE V
REPRESENTATIVE CORPUS STUDIES BY MANAGEMENT FUNCTION AND METHOD FAMILY. ENTRIES GLOSS EACH STUDY AS: STUDY SUMMARY (DATA BASIS; HEADLINE CONTRIBUTION). REFERENCES IN BRACKETS REFER TO THE NUMBERED LIST.

Function	Method family	Representative studies (data; contribution)
State estimation (SOC)	Filter-embedded learning; meta-heuristic DL	LSTM+UKF embedded estimator (laboratory cells; learned observation model in a UKF) [5]; PSO-LSTM (laboratory cells; meta-heuristically tuned recurrent SOC) [42]; GA-LSTM with adaptive Kalman filtering (laboratory cells; low error under noise) [51]
Health estimation (SOH)	Classical ML; deep learning; transfer learning	ML pipeline over 179 cells (own cell fleet; relaxed charge-curve features) [28]; deep learning without extra degradation experiments (public MATR dataset; cross-chemistry generalization) [63]; real-world vehicle SOH (fleet data; field-noise robustness) [6]
Prognostics (RUL)	Classical ML; physics-informed; RL	Early-life cycle prediction (public MATR dataset; about 9% median test error) [3]; physics-informed early-cycle RUL (multiple datasets; small-data prognosis) [73]; deep-RL prognostic model fusion (public NASA dataset; adaptive model selection) [13]
Charging control	Bayesian optimization; mechanism-aware control	Closed-loop Bayesian protocol optimization (hardware prototype; 2.5 versus 18 months to a protocol) [4]; plating-perceiving fast charging (hardware prototype; plating-free protocols) [80]; ensemble-surrogate charging optimization (laboratory cells; multi-objective strategies) [81]
Thermal management	ANN surrogate; physics-informed; optimization	ANN temperature prediction (laboratory cells; low-error thermal model) [82]; PINN heat-generation estimation (simulation; embedded thermal physics) [9]; ML-assisted BTMS design (simulation; multi-objective cooling design) [83]
Fault & safety	Shallow ML; multiphysics + ML; transfer learning	SVM fault diagnosis (hardware prototype; sensor and connection faults) [84]; multiphysics-ML runaway prediction (simulation and hardware; module-level hazard) [85]; cross-fleet runaway-diagnosis transfer (fleet data; small-sample diagnosis) [86]
Cross-cutting	Physics-informed; XAI; digital twin; LLM	EIS diagnostics PINN review (multiple studies; paradigm synthesis) [21]; uncertainty-aware early-life prediction (public MATR dataset; calibrated forecasts) [87]; LLM-based SOH estimation with uncertainty-aware field-vehicle control (field data; first published LLM estimator) [15]

force-signal deep learning catches lithium-plating-type defects at manufacturing [99]. The hazard-prediction side is increasingly model-based: multiphysics models coupled with ML predict module-level thermal-runaway progression [85], a multiphysics-informed DeepONet augments scarce experimental events with virtual data [100], and multi-source domain transfer moves thermal-runaway diagnosis across fleets with small target samples [86]. The reviews in this cluster are candid that laboratory fault diagnosis has historically failed to transfer to real vehicles [24] [23]; the fleet-data anomaly-detection studies are therefore the cluster's most valuable trend, even though label scarcity for rare faults keeps evaluation largely qualitative.

VIII. CROSS-CUTTING DIRECTIONS

A. Physics-informed learning, interpretability, and uncertainty

Physics-informed and hybrid model–data learning is the corpus's largest methodological theme (24 papers across all functions) — in state estimation [10] [101], heat-generation estimation [9], prognostics [69], and impedance-based diagnostics [21], with physics-constrained autoencoders improving health prediction over purely learned counterparts [102] and a physical-plus-data-driven fusion literature for internal-state estimation [103]. The caution is that most physics-informed studies embed simplified physics — equivalent-circuit or reduced electrochemical models — so the hybrid inherits model bias even as it gains structure, and its validation is no stronger than that of purely data-driven peers (Section 3.3). A smaller but consequential cluster makes learning systems inspectable: explainability-driven feature selection improves estimator robustness [104], interpretable ML links manufacturing coating parameters to cell capacity [105], interpretable hybrid ML untangles degradation chemistries in lithium–sulfur cells [106], and uncertainty-aware models deliver calibrated early-life predictions [87]. For safety cases only calibrated uncertainty has a route into certification; attribution supports debugging.

B. Digital twins, cloud BMS, and new sensing

System-level architecture is emerging as a research object: digital twins track individual battery aging through incremental learning [107] and couple LSTM surrogates with real-time thermal observation [108]; cloud platforms perform in-situ life prediction at fleet scale [109]; second-life workflows use impedance-based accelerated SOH estimation to triage retired packs [110], and learned battery models with numerical search estimate real-time state-of-power [111]. Non-electrical sensing broadens the measurement base — ultrasonic signatures provide a SOC channel orthogonal to voltage sensing [112] — and ML mappings from equivalent-circuit parameters to electrochemical properties give classical BMS models a physically interpretable learned layer [113]. These settings differ in latency budgets, error tolerance, and update cadence, and the field is beginning to design for those differences. Production-side mapping [114], coating interpretability [105], and defect detection [99] complete the picture: cell quality is itself a BMS input.

C. Foundation models and LLM-based management

The corpus's youngest theme is generative. The published studies to date are of two distinct kinds: a domain-specific

LLM for risk analysis of battery *energy-storage-system* incidents — station-side rather than onboard technology — [14], and HA-LLM, which genuinely crosses the estimation boundary: a knowledge-enhanced LLM that uses field vehicle data to estimate SOH with uncertainty-aware control, coordinating vehicle and cloud [15]. The honest reading of [15] matters for everything that follows: it *demonstrates* an LLM acting as an estimator with uncertainty weighting on real fleet data, so the often-quoted rule that "LLMs never estimate" is already out of date as a description of practice — what remains true is the normative claim: before any LLM-based estimator feeds actuation, it must publish calibrated intervals, structured outputs, and a latency budget, and it must not hold direct cell-level protection authority; that role belongs to physics-based envelopes and calibrated numeric estimators. Table VI makes this division of labor explicit; its right-hand column is normative guidance proposed here, not claims about the published studies.

IX. Research Gaps, Reporting Standards, and Future Directions

Cross-referencing the bibliometric analysis of Section III with the function-level reviews yields six structural gaps. (1) Validation practice: 17 primary studies use within-cell or unsupervised k-fold validation, and only 9 of all 103 demonstrate online or fleet use; standardized cross-cell, cross-chemistry, and lab-to-fleet transfer protocols should become reporting norms, as already practiced by the transfer-learning and general-framework studies [79] [78]. (2) Data monoculture: a single public dataset underlies 11 studies, NASA and CALCE sustain much of the rest, and 36 papers use private laboratory cells; federated learning over fleets, anonymized fleet-data releases, and multi-institution cycling campaigns — in the spirit of the 179-cell pipeline [28] — would diversify the evidence base, and papers should at minimum disclose split definitions, seeds, and compute budgets. (3) Edge deployment: only isolated works treat onboard compute, memory, and latency as first-class constraints [68] [115]; quantization-aware training, resource-constrained architecture search, and cloud–edge hybrid designs are the natural research program. (4) Trust: uncertainty must be calibrated rather than post-hoc [87], interpretability must bound behavior rather than merely attribute it, and hybrid physics–data architectures are the most credible certification path because part of their behavior is analytically specified. (5) Foundation-model integration: the first LLM-based estimator [15] and the risk-analysis LLM [14] suggest a division of labor — foundation models for reasoning and orchestration, conventional ML for calibrated estimation, physics models for safety envelopes — that should be formalized with grounding and latency guarantees. (6) Under-served functions: balancing is nearly untouched beyond pack-level insight [29], charging needs certifiable adaptive protocols, and thermal management remains simulation-dominated [25]; these are control problems where errors act on the battery, so closing them requires the safety-case machinery of gap 4 as much as new algorithms. Two follow-on subsections turn these gaps outward — first to what the automotive safety standards actually demand of these functions, then to how the emerging 2027–2030 research directions map onto them.

A. From corpus gaps to safety-standard requirements

The standards invoked in Section I — UN GTR 20 (battery electrical safety and thermal-propagation provisions), GB 38031 (GB/T 38031: traction-Battery abuse testing, thermal-propagation restraint, overcharge and thermal-cycling limits), and ISO 26262 (functional safety for road vehicles E/E systems) — do not name machine learning, but they define exactly what a deployed estimator or controller must not do, and Table VII translates the corpus gaps into those terms. The mapping is this review's synthesis, stated at requirement level rather than clause level: the point is not draft legal advice but that a validation protocol reported in a paper corresponds, or does not, to evidence a safety case can use. Three consequences stand out for this journal. Within-cell and k-fold validation (17 classified papers) cannot support any ISO 26262 argument, because the argument is about unseen cells and unseen duty, and the protocol does not test unseen units; thermal-propagation early-warning latency is a *timing* requirement, so fault-detection papers evaluated on curated laboratory windows under-report both the false-alarm rate and the detection-delay distribution that GTR 20's escape window would need; and charging or thermal control strategies validated only in simulation lack the hardware-in-the-loop evidence base an automotive release process expects. The reporting checklist below (Table IX) is deliberately aligned with this mapping — reporting an isolation argument and a validation environment is the minimum on which any standards conversation can start.

B. Position relative to prior reviews

The closest published competitor is a 2026 performance-centric review of ML for EV powertrains with BMS and charging scope [27]; adjacent cross-functional surveys exist for degradation and estimation methods. This review is distinguished by what those studies do not attempt, and Table VIII states the comparison: a pooled two-tier corpus with record-level screening provenance, the only systematic measurement of validation protocol strength across functions, a standards-grounded gap analysis, and a released, re-run-able analysis pipeline. Where prior reviews ask "which method works best for function X," this review asks "which claims can actually be checked — and what would it cost the reader to trust them," which is the question a scoping review is uniquely positioned to answer at corpus scale.

C. Outlook: what the next revision cycle of this literature should produce

Concretely, and tied to the gaps above, the highest-value near-term work falls in six directions. (a) *Duty-cycle-aware validation*: public cross-cell benchmarks that pair cycling data with recorded EV drive profiles, so that cross-chemistry and cross-domain generalization can be tested against transportation duty rather than laboratory schedules; (b) *federated fleet evidence*: privacy-preserving SOH/SOC studies over live

TABLE VI
SAFETY AND VERIFICATION TAXONOMY FOR FOUNDATION-MODEL COMPONENTS IN THE BMS STACK. THE RIGHT-HAND COLUMN IS NORMATIVE GUIDANCE PROPOSED BY THIS REVIEW, NOT CLAIMS ABOUT THE PUBLISHED STUDIES.

Dimension	Classical / deep ML estimators (established practice)	Published LLM studies (2025–2026)	Required guardrail (this review)
Role in the stack	Calibrated numeric estimation and detection	Stationary-BESS incident risk analysis [14]; LLM-based SOH estimation with uncertainty-aware control on field vehicle data [15]	Estimation should belong to calibrated ML; the LLM SOH estimator of [15] is the first published exception and must show calibrated intervals before any actuation-adjacent role
Grounding	Trained on domain cycling data	Domain-tuned corpora and knowledge-enhanced prompting	Retrieval over verified corpora with source citations; no ungrounded recall
Dominant failure mode	Bounded numeric error under distribution shift	Open-ended hallucinated text	Outputs restricted to structured, checkable forms; never free text into protection logic
Uncertainty	Gaussian-process / conformal intervals in a subset of studies	Not reported for incident risk analysis [14]; uncertainty-aware weighting demonstrated in the SOH estimator [15]	Calibrated intervals mandatory before any actuation-adjacent role
Latency and compute	Millisecond-class, onboard	Second-class, cloud-dependent	Offline analysis and planning only unless real-time evidence is shown
Safety authority	Bounded by physics envelopes and thresholds	Undefined in published studies	LLM components must never hold direct cell-level protection authority

TABLE VII
MAPPING BMS FUNCTIONS TO REQUIREMENTS IN UN GTR 20, GB 38031, AND ISO 26262: WHAT DEPLOYED ML COMPONENTS MUST BE ABLE TO EVIDENCE. THE MAPPING IS THIS REVIEW'S SYNTHESIS AT REQUIREMENT LEVEL (NOT CLAUSE-LEVEL LEGAL CITATION); "CORPUS STATUS" SUMMARIZES SECTIONS III–VII AND THE CLASSIFIED SUBSET.

BMS function	What the standard demands of a deployed function	Current corpus status
SOC estimation	Estimation error that can enable overcharge/overdischarge must be detected and mitigated within ISO 26262 safety goals (ASIL-rated); UN GTR 20 / GB 38031 treat overcharge as a abuse-protection requirement	43/91 primary papers use strong protocols; 10 use real vehicle data; fleet-scale SOC demonstrated once
SOH / RUL prognosis	Capacity and power fade margins underpin thermal-propagation early-warning and end-of-service decisions; degradation assumptions must hold across chemistry, duty, and environment	Accuracy claims largely on three public benchmarks (23 papers); calibrated uncertainty uncommon; fleet-proven pipelines rare
Fault detection	Thermal-propagation early warning must provide the escape window (UN GTR 20 / GB 38031 5-min provision); fault detection coverage must be argued and audited	Evaluation mostly laboratory-scale; detection-latency and false-alarm distributions rarely reported
Charging control	Overcharge, over-temperature, and thermal-cycling limits are hard constraints under GB 38031; adaptive strategies must be bounded by protection envelopes	Simulated control dominates; only a few hardware campaigns (closed-loop Bayesian optimization exemplar)
Thermal management	Temperature-gradient and propagation-containment requirements after fast-charge or abuse; cooling function must be verified on hardware	Simulation-only control loops; hardware-validated designs exist for design but not for learned policies

TABLE VIII
POSITION RELATIVE TO PRIOR REVIEWS. "PRASANNA 2026" DENOTES A 2026 PERFORMANCE-CENTRIC ML REVIEW OF EV POWERTRAINS WITH BMS AND CHARGING SCOPE; FUNCTION- AND METHOD-SPECIFIC REVIEWS ARE SUMMARIZED COLLECTIVELY. THIS REVIEW DIFFERS MOST IN THE VALIDATION MEASUREMENT, STANDARDS LINKAGE, AND RELEASED RECORD-LEVEL ARTIFACTS.

Axis	This review	Prior cross-functional / powertrain reviews
Corpus	Two-tier: 799 rule-based eligible records + 103 classified subset	Fixed study lists (e.g., 95 studies), no record-level selection provenance
Validation analysis	Systematic: protocol classes defined, per-paper assignments released, method-family contrast tested	Narrative remarks on data scarcity; protocol strength not quantified
Standards linkage	Function-to-requirement mapping vs. UN GTR 20, GB 38031, ISO 26262	Not addressed or addressed qualitatively
Recommendations	Twelve-item reporting checklist (BMS-ML-RC) + six research directions	General future-work statements
Reproducibility	Released papers.csv, screening provenance, codebook, and statistics scripts	No released record-level dataset

fleets (2% of the eligible corpus today), with the 179-cell pipeline as the template; (c) *benchmark diversification*: multi-institution cycling campaigns two orders of magnitude larger than MATR/NASA/CALCE and released with DOI-backed artifacts, retiring the benchmark-saturation pattern of Section 3.3; (d) *pack- and vehicle-level ML*: estimation validated at the integration level where UN GTR 20 and GB 38031 requirements bite, instead of cell-level evidence transplanted upward; (e) *functionally safe learning*: ML estimators whose uncertainty handling is specified against ISO 26262 safety goals (and, for capability limitations, ISO 21448/SOTIF reasoning), making certifiable charging and thermal control reachable; and (f) *honest foundation-model integration*: retrieval-grounded LLM agents with published latency and calibration budgets, capable of the coordination role while estimation authority stays with calibrated numeric models — explicitly

TABLE IX
PROPOSED BMS-ML REPORTING CHECKLIST (BMS-ML-RC), MODELED ON CONSORT/PRISMA REPORTING STANDARDS.

#	Reporting item — what must be reported
1	Dataset identity — exact name and version of every dataset (e.g., Severson-MATR, NASA, CALCE); download source
2	Cell population — number of cells, chemistry, manufacturer, format, and batch for every training and test set
3	Sensor and sampling context — measured channels, sampling rate, sensor class, temperature conditions
4	Split protocol — split type (within-cell / cross-cell / cross-chemistry / cross-domain) and isolation argument
5	Out-of-distribution evaluation — at least one protocol where test conditions differ from training
6	Baselines — at least one classical baseline alongside the proposed architecture, on identical data
7	Randomness control — seeds or repeated-run statistics for every stochastic result
8	Uncertainty — calibrated intervals or error bars for headline claims, not point estimates alone
9	Compute budget — training compute and inference latency/memory on the target platform
10	Failure analysis — at least one documented failure mode and its distribution
11	Validation environment — simulation, laboratory cycling, hardware prototype, or online/fleet
12	Artifact availability — data and code deposit (DOI-backed) or an explicit stated exemption

extending, not contradicting, the first LLM estimator [15]. A corpus-level guardrail on all six: the BMS-ML-RC checklist below is designed to be attached to a paper at submission, so the next revision cycle of this literature arrives auditable by construction.

D. A reporting checklist for machine-learning BMS studies

The most concrete artifact this review can offer is procedural. Modeled on the CONSORT and PRISMA checklists, Table IX proposes a twelve-item BMS-ML reporting checklist (BMS-ML-RC): the minimum metadata a study should disclose for its accuracy claims to be comparable, reproducible, and poolable — and, per Section 9.1, admissible in a safety discussion. Each item corresponds to a practice whose absence the corpus analysis showed to be distorting: undisclosed splits (Table III), uncalibrated uncertainty (Section 8.1), unnamed datasets and unquantified compute (Section 3.3). We suggest journals adopt it as a submission checklist and that authors attach it to their released datasets.

X. CONCLUSION

This review has examined machine learning in battery management on a two-tier evidence base: a rule-based eligible corpus of 799 journal articles from which field-level bibliometrics are measured, an in-depth classified subset of 103 studies through which validation practice is examined, and a 14-paper conference-path descriptive check. Three conclusions stand out. First, the field has grown rapidly — the eligible literature at least quadrupled between 2020 and 2023 — and, within the classified subset, method families have shifted from classical feature-based learners to deep sequence models and now physics-informed hybrids, with the first LLM-based estimator appearing in 2025–2026. Second, the center of gravity is estimation at cell level: measured across the eligible corpus, SOH, SOC, and prognostics occupy roughly half of the literature, whereas vehicle- and pack-level practice remains thin — which is precisely the level at which UN GTR 20, GB 38031, and ISO 26262 pose their requirements. Third, published accuracy progress is not matched by validation progress: only 9 of 103 classified studies demonstrate online or fleet operation, 17 use isolation-weakened splits, and 23 rest on three public benchmarks; the physics-informed validation gap is real in proportion but not statistically established (OR 0.43, 95% CI 0.16–1.14, p = 0.13), so it should be treated as an untested hypothesis rather than a settled result. The near-term agenda follows directly: standardized cross-domain validation, diversified fleet data, pack- and vehicle-level ML aligned with the standards mapping of Section IX, calibrated uncertainty, and a strict division of labor between foundation models, calibrated estimators, and physics-based safety envelopes. If the last five years were about discovering what machine learning can estimate about a battery, the next five will be decided by what it can be trusted to do — onboard, at fleet scale, across chemistries, and over full cell lifetimes.

DATA AND CODE AVAILABILITY

The complete evidentiary record of this review is released as a Zenodo deposit (DOI: https://doi.org/10.5281/zenodo.23436202): the classified corpus of 103 studies (papers.csv) with per-paper citation counts, the 799-record eligible corpus and its record-level screening provenance (screening_log.csv, 883 records), the per-paper validation-protocol audit (protocol_audit.csv), the classification codebook, and the blind 30-paper inter-coder reliability study (locked coding sheets, agreement report, itemized disagreement table). The complete search, screening, verification, and figure-generation scripts ship inside the same deposit and regenerate every figure and count in Section III; a development-form mirror is kept on GitHub.

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